r/aiwars

The true effect of AI on the environment.

Posted by u/Murky-Opposite6464 • • 0 points • 15 comments

I've written about this before, but I felt like this topic deserves a deeper, more granular analysis of the environmental impact of AI. What I intend to do is show the current environmental impact of AI, compare that to the improvements AI has proven it is capable of, and come up with an estimate of how much damage/benefit is done by AI overall.

Since I am pro-AI myself, I'm going to endeavour to be fair by going by the highest figures I can find, even if those are estimates of future use.

First, the easy part, environmental damage.

"Data centers' electricity consumption in 2026 is projected to reach 1,000 terawatts (tWh), roughly Japan's total consumption".

That combined with the fact that AI is estimated to soon be using 50% of data centre power leaves is with 500 terawatts

"AI demand for water in 2027 would be 4.2-6.6 billion cubic meters, roughly equivalent to the consumption of 'half of the United Kingdom'"

https://begin.berkeley.edu/reducing-ais-climate-impact-everything-you-always-wanted-to-know-but-were-afraid-to-ask/

I've excluded CO_2 emissions here because they are essentially a byproduct of energy use. Since the carbon intensity of electricity generation varies across regions, I chose to focus directly on the raw energy numbers, with the understanding that reductions in energy use will translate to a reduction in CO_2.

So, here are the scores.

Electricity use: 500 terawatt hours
Water use: 6.6 billion cubic meters

These are the numbers I will be comparing positive environmental impact with. And since I started writing this section first, I'm just as excited to see what the results will be as you are. :)

I'm going to be going through this field by field, sector by sector, and totalling up the results I find. To ensure total fairness, any range I am given, I will be using the lowest estimate.

Let's start with agriculture.

We are currently using 4.3 trillion cubic meters of fresh water per year, with 70% of that going to agriculture.[1] That means we are using 3 trillion cubic meters just for growing food.

I found many reports on AI controlled irrigation techniques reducing water usage by up to 30%, increasing crop yields by 20% at the same time. However, these were in third world countries that likely aren't using the most efficient non-AI methods to begin with. For a fairer comparison, I looked to a test performed in Canada that resulted in between 6.4% and 22.8% in water reduction, and a 2.3% to 4.3% increase in yield. So, going with my lowest estimate policy, that's a reduction of 6.4%, plus a negligible about for increased crop yields[2].

If we were to apply this system globally, that's a 192 billion cubic meter reduction in water use. It can reduce herbicide use too by using AI targeting, but we won't factor that in.

Well… we sure past that 6.6 billion cubic meters of water lost metric quickly.

Let’s move on to the next sector. Data centres!

The data centres that power AI, and the site you are reading this on, have been a big topic as far as increases in electrical use. However, little is said about how AI can make those data centres more efficient.

Google claimed that by using their Deepmind AI, they cut the cost of cooling their data centres by “up to” 40%, 15% of the data centres total power use.

I saw many claims of similar numbers, usually around 30%. The lowest I could find is a reduction in cooling costs from between 14% and 21%[3]. So, let’s go with 14%.

If Googles 40% claim lead to a 15% reduction overall (which seems to match other sources), a 10% reduction on cooling should be a 5.25% total reduction. So 52.5 terawatt hours.

Air conditioning!

When it comes to temperature control in general for places like offices and departments stores, I found cuts of up to 67% in various studies. The lowest I could find was 24.52%[4]. Since it had a variance of 10%, we’ll go with 14%.

To get the air conditioning power use for only large buildings, I’m simply going to subtract residential air conditioning energy use from total air conditioning use.

We get 2,100 terawatt hours for total air conditioning[5], and residential air conditioning units are 1,200 terawatt hours[6], so 900 terawatt hours for more industrial air conditioning. That’s a reduction of 126 terawatt hours.

Solar panel and wind farm optimization!

This one was fairly straightforward. AI can help position solar panels to increase energy production by 20% (I saw claims as high as 30%), and wind farm power output by 12%[7].

Solar produces 1,600 terawatt hours per year[8], leading to an increase of 320 terawatt hours. Wind generates 2,330 terawatt hours[9], so an increase of 279 terawatt hours.

So, even by the lowest metrics I could find, the increased efficiency for wind and solar farms on its own would basically offset AI’s energy consumption entirely, making everything else I’ve cited an actual reduction in global energy use. All thanks to AI.

I hope this will counter some of the misinformation I’ve been seeing out there, and put some concerns to rest. There will be challenges that come with the growth of AI, a lot will change. However, as far as the environment is concerned, these changes will be for the better.

1: https://reports.weforum.org/docs/WEF_Water_Futures_Mobilizing_Multi_Stakeholder_Action_for_Resilience_2025.pdf

2: https://arxiv.org/abs/2306.08715

3: https://arxiv.org/abs/2501.15085

4: https://www.researchgate.net/figure/The-average-energy-savings-of-the-24-cases-and-the-maximum-energy-savings-achieved-by_fig6_331608687

5: https://ourworldindata.org/air-conditioning-causes-around-greenhouse-gas-emissions-will-change-future

6: https://www.nature.com/articles/s41467-024-52028-8

7: https://ojs.stanford.edu/ojs/index.php/intersect/article/view/3541?utm_source=chatgpt.com

8: https://www.iea.org/energy-system/renewables/solar-pv

9: https://www.iea.org/energy-system/renewables/wind

Fun little bonus facts. The positive effects of AI on carbon emissions in just 3 industries (meat and dairy, power, and light road vehicles) would completely negate all damage done by increased power consumption from expanding AI use.

https://www.nature.com/articles/s44168-025-00252-3

UPS optimized their deliveries with AI, saving 10 million gallons of fuel.

https://www.bestpractice.ai/ai-case-study-best-practice/ups_saves_over_10_million_gallons_of_fuel_and_up_to_%24400m_in_costs_annually_with_advanced_telematics_and_ana

Pittsburg used AI controlled stoplights to cut emissions from cars by 21%, and reduced total driving time by 26%.

https://govlaunch.com/stories/pittsburgh-pa-reduces-traffic-congestion-with-ai

Comments

u/Aggravating-Play5388 • 1 points • 2025-11-14 16:32 UTC

Appreciate the effort to use conservative (high) cost estimates and low benefit estimates. A few nuances I've seen matter when you scale this kind of analysis:

* Projections vs. realized use: 2026–2027 numbers are useful stress tests, but actual impact will hinge on adoption curves, datacenter mix (efficiency, cooling type), and where/when workloads run on the grid.
* System boundaries: stacking percent gains across sectors can double-count if improvements aren't strictly AI-dependent or if only a fraction of sites adopt them.
* Rebound effects: efficiency wins (AC, routing, cooling) can be partially offset by increased usage unless pricing/policy keeps demand in check.
* Water/energy are local: siting and cooling make order-of-magnitude differences, so global totals can hide regional constraints.

I'm pro-AI and agree the net can be positive if we pair "reduce first" with "restore too." That's why I started Earthly Insight as a small alternative in the AI space: we keep the product text-first (lighter than image/video workloads), donate a share of revenue to vetted rewilding projects as a complement, not a substitute to decarbonizing compute, and we're raising to build a more efficient, domain-tuned model so we can deliver the same utility with fewer tokens and watts. Happy to share assumptions and let folks pressure-test our transparency if that's helpful.

If you want a shorter version Great analysis. My only flags are adoption/double-counting and rebound effects—site-level gains don't always translate 1:1 globally. We started Earthly Insight (www.earthlyinsight.com) to offer a lighter alternative: text-first by design, funding vetted rewilding as a complement to cutting compute, and working on a more efficient domain-tuned model. Open to tough questions on our approach.

u/Aggravating-Play5388 • 1 points • 2025-11-12 19:34 UTC

Two quick nuances I'd add as someone building in this space:

\-Text-first vs. heavier media: A chat reply is a single forward pass over tokens, whereas image/video generation (e.g., diffusion) runs many iterative steps on much larger tensors, so per-output compute, energy, and water are materially higher for images/video than for text. Designing products to stay text-first helps keep the footprint down.

\-Offsets vs. restoration: Rewilding/restoration can store carbon and restore biodiversity, but it's a complement to decarbonizing compute and grids - not a substitute. The honest path is "reduce first, restore too."

For transparency: I run a small, text-first AI project called Earthly Insight. We donate a share of revenue to vetted rewilding projects, and we're currently seeking funding to build a more efficient, domain-tuned LLM (distilled/compressed and scheduled on cleaner grids) so we can deliver the same utility with fewer tokens and watts. Happy to share our assumptions and let folks pressure-test our transparency.

u/Turbulent-Gur1314 • 1 points • 2025-11-11 02:20 UTC

But what about those living in cities with data centers that have been built due to the increase in demand for AI? Not only has the amount they have to pay for utilities gone up, but clean water is increasingly hard for them to come by.

u/LivingInMyBubble1999 • 2 points • 2025-10-02 02:17 UTC

This doesn't even take in account innovations that are coming precisely because of AI scaling.
Extropic computer chips consume 1000 to 10,000 less electricity , while cutting 80-90 percent of cooling needs. This won't even exist if AI demand wasn't increasing faster than compute capacity.
We can't even begin to solve Fusion without AI.
If Helion succeeds in 2028, credit goes completely to AI since it's a very important component of how their system works.

u/Mysterious_Ice_3722 • 2 points • 2025-09-29 21:50 UTC

Fantastic, I really appreciate this post.

The argument that AI destroys the environment needs to be put to rest. I for one wouldn't even consider myself pro-AI. But the fact is, AI is not environmental threat, quite the opposite. Its an opportunistic tool that can be used to better the environment. There's alot more things our there that are more of a threat to environment then AI. Keep up the good work 👏

> **u/Murky-Opposite6464** • 1 points • 2025-09-29 23:31 UTC
>
> Thank you. :)

u/Most-Structure-9116 • 1 points • 2025-09-29 18:53 UTC

Yeah I never believed AI was really going to destroy the environment, we are already really good at doing that. It was the other aspects I took issue with.

u/RouxMango80 • 1 points • 2025-09-29 14:26 UTC

I'll make you a deal: I'll stop eating beef and using AI, and you stop eating beef and using AI. Sound good?

> **u/Raffino_Sky** • 0 points • 2025-10-01 13:33 UTC
>
> If you stop eating beef, there will be leftovers (unless you hunt for/slaughter your beef yourself?).
>
> So I'll eat that to avoid loss of perfectly fine beef, and I keep using AI for usefull stuff. Deal?

> **u/Murky-Opposite6464** • 1 points • 2025-09-29 23:26 UTC
>
> You are aware this is a pro-AI post, right?… or did you just read the title and make assumptions?
>
> > **u/RouxMango80** • 1 points • 2025-09-30 00:58 UTC
> >
> > Of course you're defending AI from all those hypocrite carnivores. I, too, have noticed that industries get away with exploitation because other, different industries get away with exploitation. Do you suppose it's a trend?

u/gnolex • 2 points • 2025-09-29 13:00 UTC

The problem here is that you're conflating generative AI with AI used for control systems. These two are vastly different technologies and using both for analyzing effects of AI on environment is very flawed.

AI used for control systems are optimized for size and efficiency. They can automatically figure out the most efficient way of controlling the system and they can adapt to changes dynamically. They are usually designed to be small, often a very specific number of layers and neurons in each layer, mainly to simplify the system but also to protect it from overfitting. I taught mechatronics students about it a few years ago, they were given a simulation of a servomotor used in some industrial machine and told to teach neural network to control it based on input. Energy usage for training is minimal, you can use 10+ years old machine to train it in minutes. Then you can implement the system directly into a chip, program FPGA or use AI accelerator for it. We've been using this technology for years and it was never a problem. And yes, proper usage of AI in control systems tend to reduce CO2 emission, as well as usage of water and energy. That's one of the main selling points of the technology.

Generative AI is a completely different beast. You design the system to be as large as possible in order for it to fit as much information as possible. It was never designed with efficiency in mind so it has massive issues with overfitting and pattern duplication. Generative AI learns by brute forcing extraction of patterns based on a gigantic amount of information without having any sort of context beyond labels provided among the data. The growth of these systems is much faster than growth of their capabilities. And the only reason large AI systems like current versions of ChatGPT are even possible is because they use large amounts of memory and computational power of GPUs to power them. Now they have to build data centers to power them as they are growing too large to handle on individual level.

Back when I was a PhD candidate, I was helping my colleague on his work on a denoiser for reconstruction of text with missing information. The whole point was to figure out AI system capable of helping with text reconstruction that would fit on a simple computer or maybe even a dedicated processing chip. It was never about using as much GPU resources as we could. But after denoisers and later stable diffusion were used to make generative AI, the whole optimization part went out the window as hype took over and now it's all about building the largest AI model possible that will take as much resources as it needs to. Environmental impact will only get worse and efficiency generated by other forms of AI will not be able to compensate for growing hunger for resources by generative AI. Exponential growth of generative AI is not sustainable in the long run.

> **u/Murky-Opposite6464** • 1 points • 2025-09-29 23:19 UTC
>
> I did not conflate the two, I just didn’t exclude generative AI from the calculations, which is entirely fair considering the damages of AI are calculated in the same fashion.
>
> The paper I cited already explains that the overall harm to the environment will be completely offset by just three major industries adopting AI, which they will, as energy efficiency equals saved money, and no company is going to leave that money on the table.

And the idea that generative AI isn't becoming more efficient may have been true in the past, but there have been substantial advances in recent years.

"the inference cost for a system performing at the level of GPT-3.5 dropped over 280-fold between November 2022 and October 2024. At the hardware level, costs have declined by 30% annually, while energy efficiency has improved by 40% each year."

https://hai.stanford.edu/ai-index/2025-ai-index-report?utm_source=chatgpt.com

And running large language models can reduce their carbon output by 40% through quantization.

https://arxiv.org/abs/2504.06307

You are also forgetting that generative AI will be used to design more efficient systems. Designing more efficient microchips will utilize generative AI. Designing more efficient engines will use generative AI. You can't just disregard generative AI when it comes to increased efficiency, and reduced resource consumption.

u/AppearanceHeavy6724 • 2 points • 2025-09-29 18:24 UTC

> Generative AI learns by brute forcing extraction of patterns based on a gigantic amount of information without having any sort of context beyond labels provided among the data.

LLMs are trained with unlabeled data.

> it has massive issues with overfitting and pattern duplication.

"overfitting" is so misused word, it means almost nothing these days. I have no idea what is "pattern duplication".

> often a very specific number of layers and neurons in each layer, mainly to simplify the system but also to protect it from overfitting.

LLMs have "specific" number of weights in each layer too.

> Generative AI learns by brute forcing extraction of patterns based on a gigantic amount of information without having any sort of context beyond labels provided among the data.

Any neural network is trained by backprop exactly same way. Loads of training data needed for good performance.

> Environmental impact will only get worse and efficiency generated by other forms of AI will not be able to compensate for growing hunger for resources by generative AI.

A single prompt to Gemini consumes 0.25 Wh, something I run on my personal LLM on my PC it is about 2 Wh. Negligible numbers on per person scale.

u/jay-ff • 1 points • 2025-09-29 21:14 UTC

Just on that last part. Personal numbers or numbers per query are environmentally irrelevant. The total number is what counts and that is enormous and it will get as big as possible, constrained only by the available resources. The demand is currently much bigger than the supply when it comes to compute power.

u/AppearanceHeavy6724 • 1 points • 2025-09-30 10:30 UTC

>Personal numbers or numbers per query are environmentally irrelevant. The total number is what counts and that is enormous and it will get as big as possible, constrained only by the available resources.

This is a ridiculous take. The total and per person environmental footprint of consuming meat or driving car is so massively higher than for AI inference it is idiotic to even bring that up.

> The demand is currently much bigger than the supply when it comes to compute power.

This can be said about any any type of goods. Demand for luxury housing, for iPhones, for meat - is all currently much bigger than the supply. Keep in mind that dollar spent on AI is dollar not spent on gasoline or a burger. People do not have infinite pockets.

u/jay-ff • 1 points • 2025-09-30 11:39 UTC

>This is a ridiculous take. The total and per person environmental footprint of consuming meat or driving car is so massively higher than for AI inference it is idiotic to even bring that up.

Numerous things can have a massive energy consumption. Always just looking at the top entries in that list is short-sighted in my opinion, especially because the methods to mitigating those problems are different. Also: My point was about single query energy consumption, which is irrelevant because whether you use AI a bit more or a bit less doesn't affect the environment. The total number is what counts and if that is high despite individual query cost being low, that means they people are using it to a degree that adds up to a large number. I don’t think that this is ridiculous. People are way to hung up on the personal footprint way of seeing things. Comparing one query to chatGPT or Gemini to say writing an essay or painting a picture are the definition of apples to oranges because all the secondary parameters for the environmental impact are ignored. If you feel that the power-consumption of AI as a whole isn’t high enough to be concerning, that’s a fair take. But the single query energy consumption is not what counts here.

>This can be said about any any type of goods. Demand for luxury housing, for iPhones, for meat - is all currently much bigger than the supply.

Thats just not true with the exception of luxury housing maybe. You can pick up an iPhone or as much meat as you want right now. The production is more or less matched with demand, except special cases like whenever a new PlayStation comes out or suddenly people start buying up graphics cards for bitcoin mining. If the supply was much bigger than the demand you wouldn’t see every second person using an iPhone and basically everybody eating meat daily (at least in developed countries) at fairly modest prices.

AI hast reached the saturation yet. I don’t have any concrete data about the individual segments are driving the demand from the consumer side (my guess is probably mostly software people) but ai companies want to deploy the biggest models they can and the computing power used for AI hast been growing extremely steeply (see every first graph in presentations from NVIDIA or openAI). In my opinion this is relevant because it means that the energy consumption will very likely go up from here. How much is of course not clear. If AI proves to be super useful, probably by a lot.

>Keep in mind that dollar spent on AI is dollar not spent on gasoline or a burger. People do not have infinite pockets.

If that were true, you could measure it but since the demand for those things depends on different factors, I highly doubt it. People are not forced to choose between AI and meat or cars. People don’t even have to spend all their money at all.

u/AppearanceHeavy6724 • 1 points • 2025-09-30 11:49 UTC

> You can pick up an iPhone or as much meat as you want right now.

First world thinking. I live in developing world buddy, and trust me iPhone and beef is in way higher demand than compute here.

u/Zorothegallade • 3 points • 2025-09-29 10:32 UTC

In layman's terms, we are investing that water and power on the long terme.

u/lastberserker • 10 points • 2025-09-29 09:59 UTC

Copying from a comment I posted earlier:

Here is Wisconsin Public Radio claiming that the new AI data center built there will consume **8.4 million** gallons of water: https://www.wpr.org/news/microsoft-data-centers-8-million-gallons-water-each-year Huge number, right? One might even say unprecedented!

That's enough water for about 15,000 Big Macs. Enough to feed a dozen adults for a year, if they eat nothing else.

Or you can grow about 5 tons of almonds.

Perspective is everything.

u/waddleship • 1 points • 2025-09-29 11:21 UTC

Yeah that’s not how data analysis works bro

u/IgnitesTheDarkness • 7 points • 2025-09-29 11:05 UTC

sounds a lot until you figure that there are millions of people in Wisconsin and they are eating meat and driving cars and doing any number of other things that damage the environment om a daily basis. AI is still a drop in the bucket especially if it can do things to mitigate other environmental harms.